Unit 1.0 — Introduction to Indian Knowledge Systems

1.1 Overview of IKS

Indian Knowledge System (IKS) refers to the vast, ancient, and continuing body of knowledge developed in the Indian subcontinent over thousands of years, encompassing science, mathematics, philosophy, medicine, architecture, arts, and ethics. NEP 2020 emphasizes integrating IKS into modern technical education to help learners appreciate India’s intellectual heritage and apply its wisdom to contemporary problems.

1.2 Organization of IKS — चतुर्दश विद्यास्थानं (The Fourteen Branches of Knowledge)

Traditional Indian knowledge was systematically organized into **fourteen (चतुर्दश) Vidyāsthānas (branches/seats of learning)**, broadly comprising the four Vedas, six Vedāngas (auxiliary disciplines), and four Upāngas (Purāna, Mimāmsā, Nyāya, Dharmashastra) — covering spiritual, scientific, linguistic, and social domains of knowledge in an integrated manner.

1.3 Conception and Constitution of Knowledge in Indian Tradition

In Indian tradition, knowledge (Vidya/Jñāna) was viewed holistically — not separated into “science” and “spirituality” as in the modern Western framework, but seen as an integrated pursuit of truth (Satya) through direct experience (Anubhava), logical reasoning (Tarka), and scriptural authority (Shabda/Āgama). Knowledge was considered a means to liberation (Moksha) as well as practical welfare (Artha).

1.4 The Oral Tradition

Much of ancient Indian knowledge was preserved and transmitted **orally** across generations through structured memorization techniques (e.g., Ghana-patha, Krama-patha for Vedic recitation), ensuring extremely high accuracy of transmission over centuries, even before the widespread use of written texts. The Guru-Shishya (teacher-disciple) tradition played a central role in this oral transmission.

1.5 Models and Strategies of IKS

Indian knowledge transmission relied on models such as: - **Guru-Shishya Parampara:** Direct mentorship between teacher and student. - **Sutra style:** Knowledge condensed into short, precise aphorisms (sutras) for easy memorization, later elaborated through commentaries (Bhashya). - **Discourse and dialogue (Samvada):** Many texts (e.g., Bhagavad Gita) are structured as a dialogue/discussion, encouraging questioning and reasoning rather than passive acceptance.

Unit 2.0 — Overview of IKS Domains and Relevance in Current Technical Education System

2.1 The Vedas as the Basis of IKS

The **four Vedas** — Rigveda, Yajurveda, Samaveda, and Atharvaveda — are the foundational texts of Indian knowledge, containing hymns, rituals, philosophical thought, and early observations of nature, considered the source from which other branches of IKS (Vedāngas) developed.

2.2 Overview of the Six Vedāngas

The **six Vedāngas** (limbs of the Veda) are auxiliary disciplines developed to correctly understand and apply Vedic knowledge:

Vedānga	Focus Area
Shiksha	Phonetics and pronunciation
Kalpa	Rituals and procedures
Vyakarana	Grammar
Nirukta	Etymology
Chhandas	Metrics/prosody
Jyotisha	Astronomy/astrology, calendar calculations

2.3 Relevance of IKS Domains in Present Technical Education

- **Arthashastra (Indian economics and political systems):** Kautilya's Arthashastra discusses statecraft, taxation, governance, and economic policy — relevant to modern management, public administration, and economics.
- **Ganita and Jyamiti (Indian Mathematics, Astronomy and Geometry):** Indian scholars (Aryabhata, Brahmagupta, Bhaskara) contributed concepts like zero, decimal system, algebra, and trigonometry, foundational to modern mathematics and computer science.
- **Rasayana (Indian Chemical Sciences):** Ancient Indian alchemy and metallurgy (e.g., rust-resistant Iron Pillar, Ayurvedic chemical processes) relevant to modern materials science and chemistry.
- **Ayurveda (Indian Biological Sciences / Diet & Nutrition):** A holistic medical system focusing on balance of body, mind, and diet, relevant to modern healthcare, nutrition science, and biotechnology.
- **Jyotish Vidya (Observational astronomy and calendar systems):** Accurate astronomical observations led to precise calendar systems and predictions of celestial events, relevant to modern astronomy and satellite-based applications.
- **Prakriti Vidya (Terrestrial/Material Sciences, Ecology, Atmospheric Sciences):** Ancient understanding of nature, ecological balance, and sustainability, relevant to modern environmental engineering and climate studies.
- **Vastu Vidya (Aesthetics, Iconography, Architecture):** Principles of design and construction based on natural elements, orientation, and aesthetics, relevant to modern architecture and civil engineering.
- **Nyaya Shastra (Social Ethics, Logic and Law):** A system of logic and epistemology, relevant to modern computer science (logical reasoning, AI), philosophy, and judicial systems.
- **Shilpa and Natya Shastra (Performing and Fine Arts):** Classical treatises on art, sculpture, dance, and drama, relevant to modern design, animation, and media studies.
- **Sankhya and Yoga Darshana (Psychology, Yoga, Consciousness studies):** Philosophical schools dealing with the nature of mind and consciousness, relevant to modern psychology and cognitive science.
- **Vrikshayurveda (Plant Science / Sustainable agriculture):** Ancient Indian texts on plant care, agriculture, and food preservation, relevant to modern agricultural and food technology.

Significance for engineering students: Correlating these domains with engineering education encourages multidisciplinary thinking, promotes sustainable and ethical solutions rooted in Indian heritage, and builds an appreciation for indigenous innovation alongside modern technology.